Frozen-Encoder Guidance for Diffusion Text Generation on a Semantic Map

point2text contributors - loopmath

Experimental technical report - 29 October 2026

This implementation steers a frozen local diffusion language model toward a clicked point on a fixed embedding map. It adds encoder-input gradients inside denoising while retaining the outer generate-and-verify search. Real probes confirm that guidance changes native sampling, decoded text, and its measured location. They do not establish a general quality or efficiency advantage.

1 Target and native runtime

The fixed map uses 100 authored paragraphs and normalized 384-dimensional all-MiniLM-L6-v2 embeddings [1]. Canonical scoring means ordinary tokenization, frozen encoder inference, attention-mask mean pooling, L2 normalization, and saved unwhitened PCA; inputs over 256 encoder tokens are rejected.

$$\pi(e) = V^T(e - \mu), \qquad d(x, y^*) = \|\pi(E(x)) - y^*\|_2.$$

The pinned four-bit DiffusionGemma checkpoint [2] runs through MLX-VLM 0.7.11 [3]: 128 canvas positions, 262,144 vocabulary entries, and at most 32 passes. Prompt conditioning is cached within each trajectory; the exact checkpoint revision is recorded in run artifacts.

The entropy-bound sampler can revise the whole canvas and stop early on confidence/stability. Its temperature-adjusted logits are detached before guidance. Revised logits drive sampling, acceptance, self-conditioning, and stopping; the generator receives no backward pass.

2 Sparse encoder surrogate

At a guided pass, z0 denotes the native clean-token logits. Each position's top-32 support S_i stays fixed during that pass's inner updates; only its selected logits z are optimized. The conditional distribution is q_i = softmax(z_i / tau).

For generator token v, isolated decoding and encoder re-tokenization produce token list T_v. The bridge averages MiniLM's input lookup vectors W_E, not sentence embeddings. With validity mask a_v, represented rows renormalize probabilities:

$$B_v = \frac{1}{|T_v|} \sum_{k \in T_v} W_E[k], \qquad u_i = \frac{\sum_{v \in S_i} q_{iv} a_v B_v}{\sum_{v \in S_i} q_{iv} a_v}.$$

The sequence [CLS, u_1, ..., u_n, SEP] passes through the frozen MiniLM body, masked mean pooling, L2 normalization, and PCA. Unrepresented rows use fixed-token fallbacks and masks. The editable-position loss is:

$$L(z) = \|\pi(E_{\text{soft}}(z)) - y^*\|_2^2 + \frac{\lambda}{|M|} \sum_{i \in M} \mathrm{KL}(q_i \| q_i^0).$$

The original proposal q_i^0 is fixed across inner updates. Gradients follow map loss through pooling, MiniLM inputs, mixtures, and softmax into selected logits; neither hard decoding nor the generator is differentiated. The first KL gradient is zero at $q_i = q_i^0$.

Soft and canonical inputs differ concretely: two encoder subwords become one averaged vector at one position here; normal encoding retains both positions and their contextual attention. Joining decoded surfaces can also change tokenization. Top-k omissions and final EOS clipping add discrepancies. Thus lower soft loss need not lower completed-text distance.

3 Outer search and hard verification

For each of N independent seeds, reuse the four-neighbor prompt:

```
canvas <- initialize(seed); self_condition <- none
cache <- prefill(prompt)
for native pass t = 1,...,T:
  z0 <- frozen_generator(canvas, cache, self_condition)
  z <- detached_copy(z0)
  if guidance_scheduled(t):
    S <- top32(z0); q0 <- softmax(z0[S] / tau)
    repeat m times:
      g <- gradient_z[S] L(z[S]; S, q0, target)
      z[S] <- bounded_gradient_update(z[S], g, z0[S])
  canvas, self_condition <- native_sampler(z)
  stop if native confidence/stability criterion holds
x <- decode(final_tokens)
d <- canonical_distance(x, target)
```

Updates normalize and clip the global gradient direction, take a scaled descent step, then cap cumulative changes from z0. Fixed m (default two) does not test convergence. Canvas and self-conditioning carry influence forward; each pass generates fresh z0, not persistent optimized logits.

Native confidence/stability stopping does not test target epsilon. Canonical scoring of completed, normally re-tokenized text selects the best of N; it never feeds a revision or next candidate. No epsilon = 1e-4 stopping rule is implemented.

4 What the probes establish

On an M4 Max with 128 GiB memory, paired probes used the same corpus target, neighborhood context, seed 42, checkpoint, and native generation budget.

Profile	Canonical distance	Observed intervention
Unguided	0.07483	Nine native passes
Strong late guidance	0.09050	Four guided passes; text changed
Strong early guidance	0.02411	Thirteen guided passes; text changed

These profiles differ on one development target. Early guidance used two updates at all 13 passes with maximum logit change four; late guidance began at pass six with a tighter bound. General benefit and text quality were not established.

5 Computational cost and next gate

Candidates run sequentially and repeat prompt prefill. Top-k reduces bridge work, not generator forward cost. The early profile adds 26 encoder forward/backward pairs plus synchronization; more inner updates add encoder work. Decoder compilation is disabled.

Cold/warm differences show no speedup. Compare guidance with more unguided samples under equal wall time across targets, measuring canonical error, quality, diversity, and memory. Hard/soft estimators [4] remain a future comparison. Low map distance alone is not semantic equivalence.

References

[1] Sentence Transformers. all-MiniLM-L6-v2. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[2] Google. DiffusionGemma model card. https://ai.google.dev/gemma/docs/diffusiongemma/model_card

[3] MLX-VLM. Native inference implementation; version 0.7.11 used here. https://github.com/Blaizzy/mlx-vlm

[4] Tejaswi et al. Entropy Aware Reward Guidance for Diffusion Language Model Alignment. v2, 2026. https://arxiv.org/abs/2602.05000